

NAVIGATING THE 1W ENVELOPE: AREA-ENERGY TRADE-OFFS OF SCALABLE RISC-V SYSTOLIC ARRAYS IN SKY130

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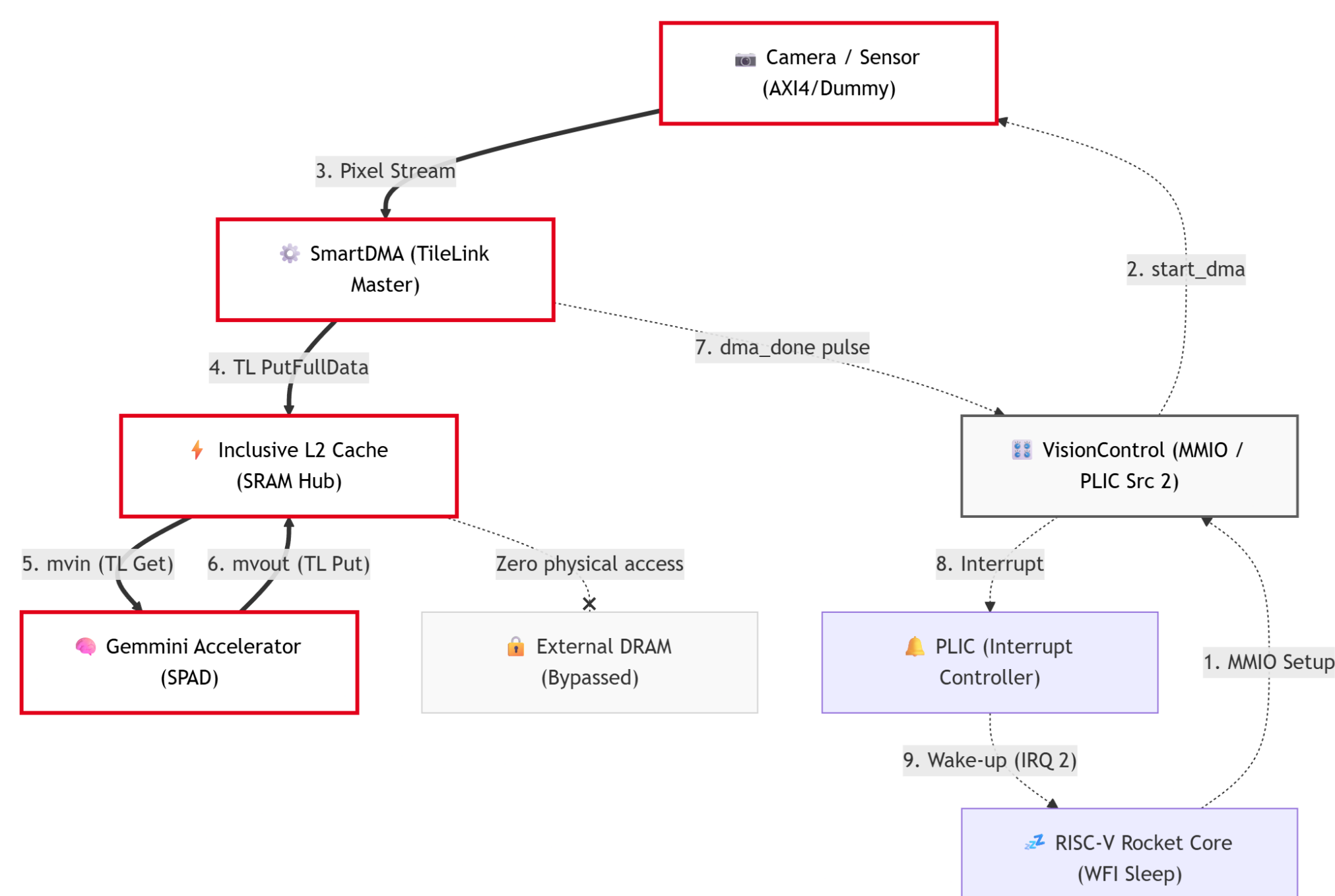
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1. Motivation & Context

- **Edge-AI for MAVs:** Deploying high-performance AI inference (like real-time object detection or high-res vision) on micro-aerial vehicles requires a strict **1W power budget** to preserve flight stability.
- **Sovereign Technology:** Due to open-source hardware sovereignty trends, mature nodes like **SkyWater 130nm** are increasingly relevant compared to expensive FinFET nodes.
- **The Challenge:** Systolic arrays maximize tensor efficiency, but scaling them in a 130nm node leads to dramatic area-induced leakage and routing congestion.

2. Hardware Architecture & L2-Handoff

- **Design Space Exploration:** Vertical evaluation of the RISC-V **Gemmini** accelerator (Chipyard), scaling from 8x8 to 32x32 systolic meshes.
- **Hierarchical “Tile-and-Stitch”:** Developed to mitigate the “area explosion” of synthesized register files in 130nm using verified 128x32 OpenRAM macros.
- **Zero-Copy L2-Handoff (New):** A custom TileLink DMA (**SmartDMA**) streams sensor data directly into the Gemmini scratchpad via the inclusive L2 cache. **Zero external DRAM transactions** occur during the burst.
- **WFI Power Secret:** The CPU triggers the DMA and enters the **wfi** (Wait-for-Interrupt) sleep state. It is woken up via a diplomatic PLIC interrupt exactly once per frame, massively reducing active power.

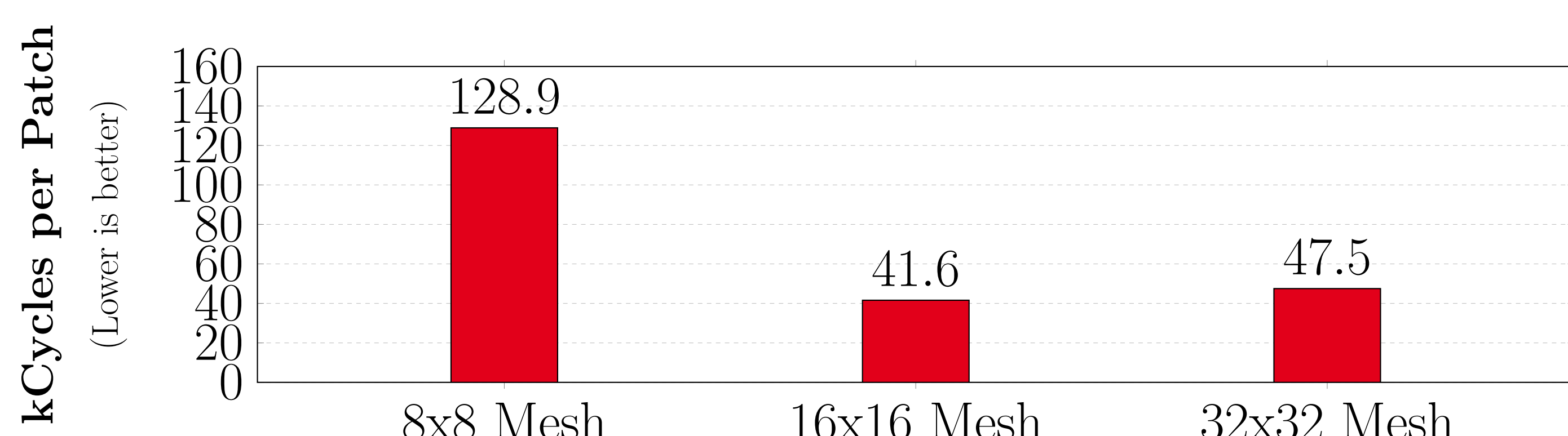


3. Software & Quantization Pipeline

- **Toolchain:** **ONNX Runtime** and **Apache TVM** for Post-Training Quantization (PTQ) to INT8.
- **Memory Reduction:** Model footprint reduced by **74.75%**, enabling weights to fit within the on-chip memory hierarchy with an mAP loss **below 2.0%**.

4. Experimental Results (VisDrone & High-Res PoC)

- **Initial VisDrone Evaluation:** Real-world evaluation on 20 images against a RISC-V Rocket CPU baseline showed the **16x16 mesh** as the sweet spot (2352x speedup at 428 mm²).
- **High-Res PoC (640x640 Proxy):** Our latest bare-metal PoC runs an end-to-end evaluation for high-resolution vision patches.
- **Performance Scaling:** The chart below reveals that the **16x16 mesh** achieves the lowest latency, avoiding the architectural bottlenecks of larger arrays:

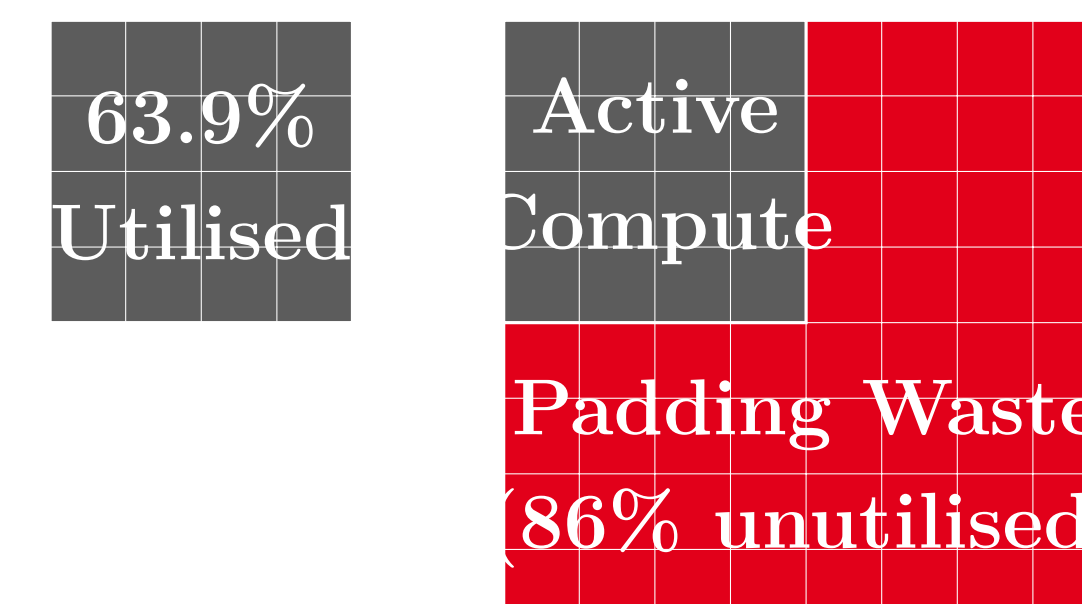


5. Key Insights: The Padding-Waste Phenomenon

- **Channel-Padding-Waste at 32x32:** While a 32x32 mesh doubles the compute resources, it forces aggressive padding for narrow early layers (e.g., padding 16 out-channels to 32).
- **The Performance Collapse:** Utilisation crashes to 14.0%, making the 32x32 array actually **14% slower** (47.5k vs 41.6k cycles) than the 16x16 mesh for this workload.
- **Energy Efficiency Impact:** Unutilised Processing Elements (PEs) continue to toggle and dissipate clock-tree power. This results in severe energy overhead, destroying the energy efficiency benefits (GOPs/W) of spatial scaling.
- **Array Sizing Rules:** Spatial edge accelerators must align with typical network channel dimensions. A 16x16 mesh represents the optimal architectural “sweet spot” for balancing throughput and structural utilization.

16x16 Mesh 32x32 Mesh

(The Sweet Spot) (Scaling Bottleneck)



6. Conclusion & Path to 1W Target (Future Work)

- **Brute-Force Baseline:** Reaching target inference rates (15,360 patches/s) with a single 16x16 core in 130nm draws **≈ 9.0W** (extrapolated).
- **Hardware-Software Co-Design:** Pure hardware scaling fails due to spatial underutilization. Squeezing advanced vision models into a tight 1W envelope demands structured model pruning that strictly matches array dimensions.
- **The 1W Roadmap:** Future work will focus on three key pillars:
 - **Parallel 16x16 Clusters:** Distributing workloads to lower individual clock frequencies and voltage domains.
 - **Structured Sparsity:** Exploiting zero-skipping in hardware to drop weight data volume down to 0.05.
 - **Dynamic Scaling:** Duty-cycling the accelerator domain (e.g., targeting 8 fps to achieve **≈ 1.12W** active power).
- **Mature Node Relevance:** This exploration proves that with intelligent zero-copy cache handoff, mature open-source nodes like Sky130 remain highly viable for sovereign, low-cost edge-AI hardware.

7. References & Project Links

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